CSE 270 Sports Analytics

Regresion to the mean

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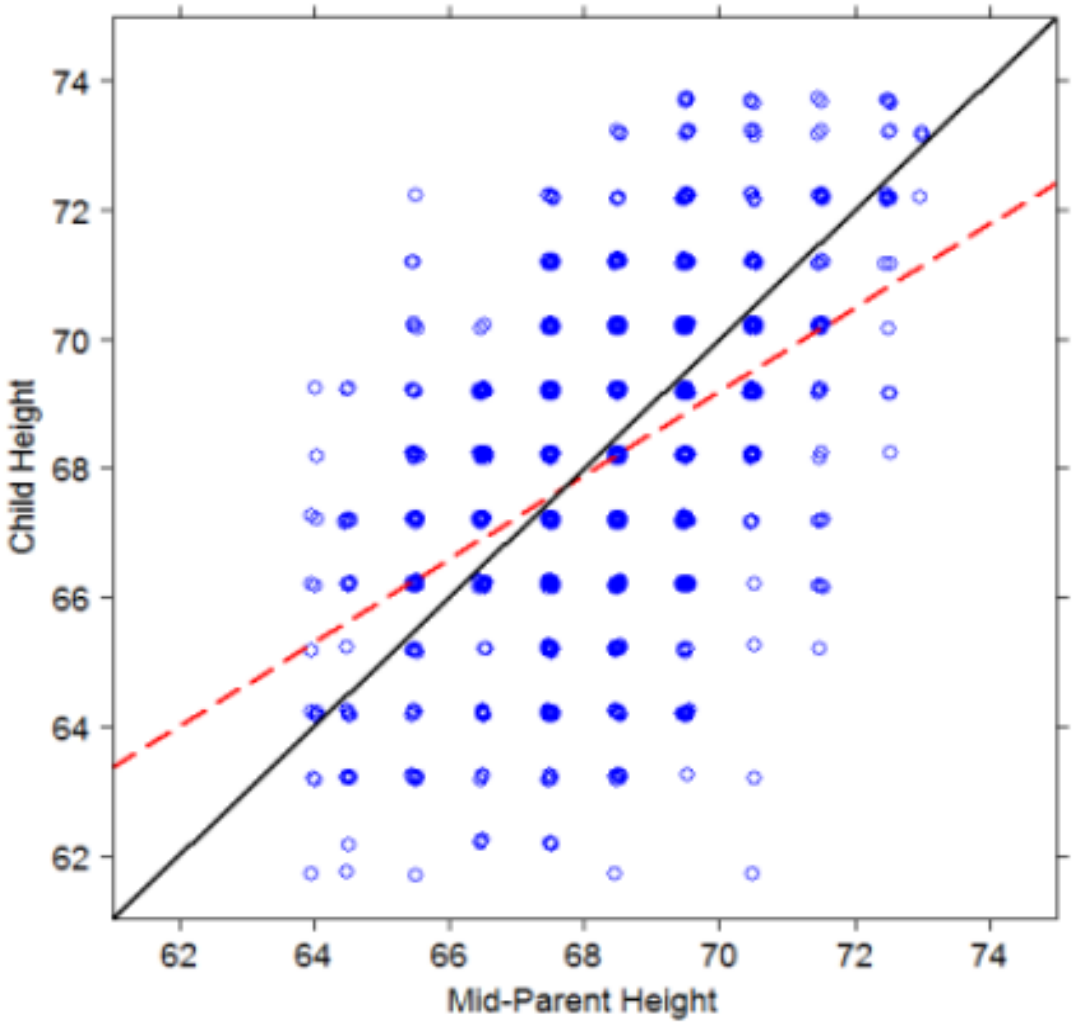
Regression to the mean

Libraries used

```
library(knitr)
library(dplyr)
library(ggplot2)
library(measures)
library(odds.converter)
library(SportsAnalytics270)
```

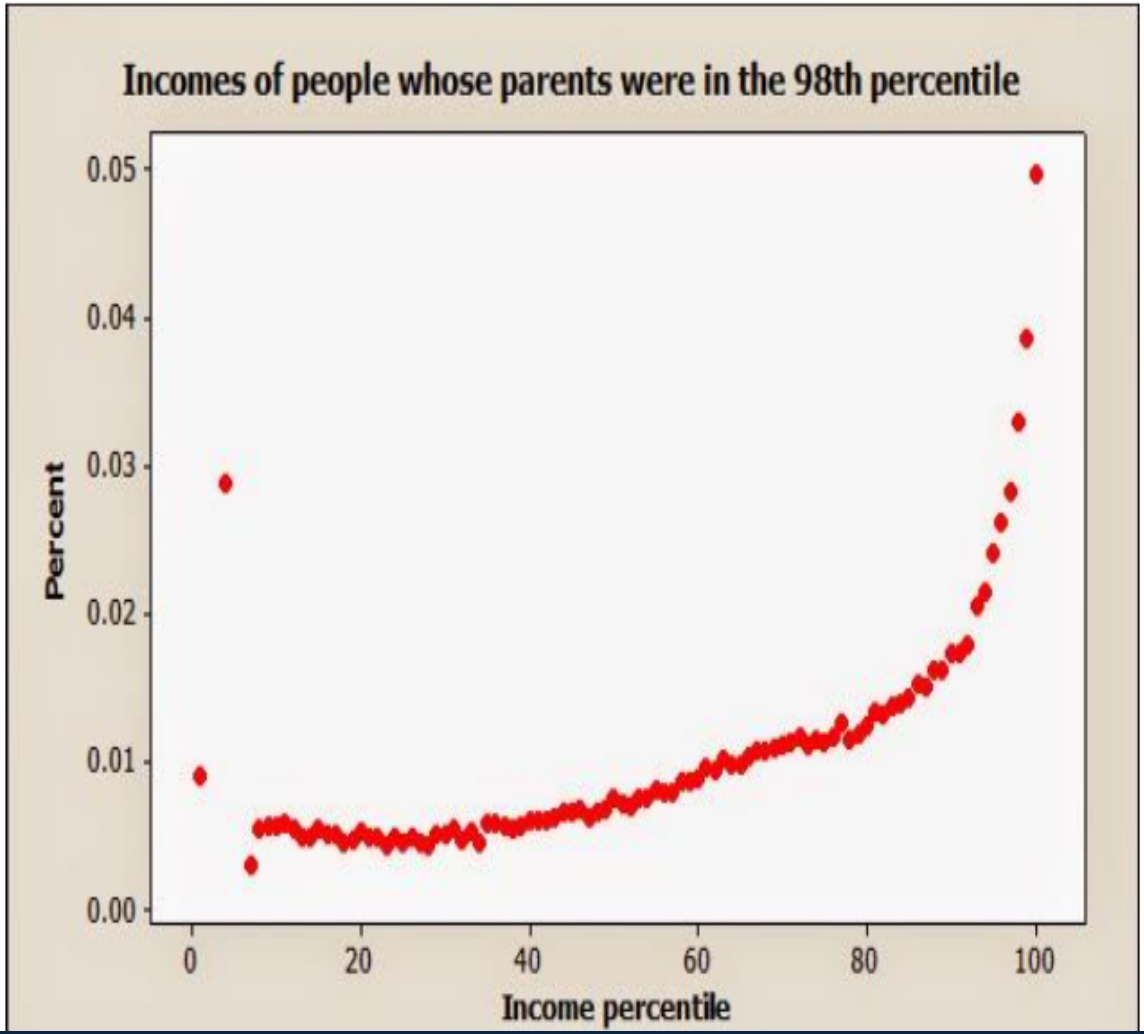
Regression to the mean: Height

Sir Francis Galton proved that the kids of shorter parents have higher height than their parents, and kids of taller parents have shorter height than their parents



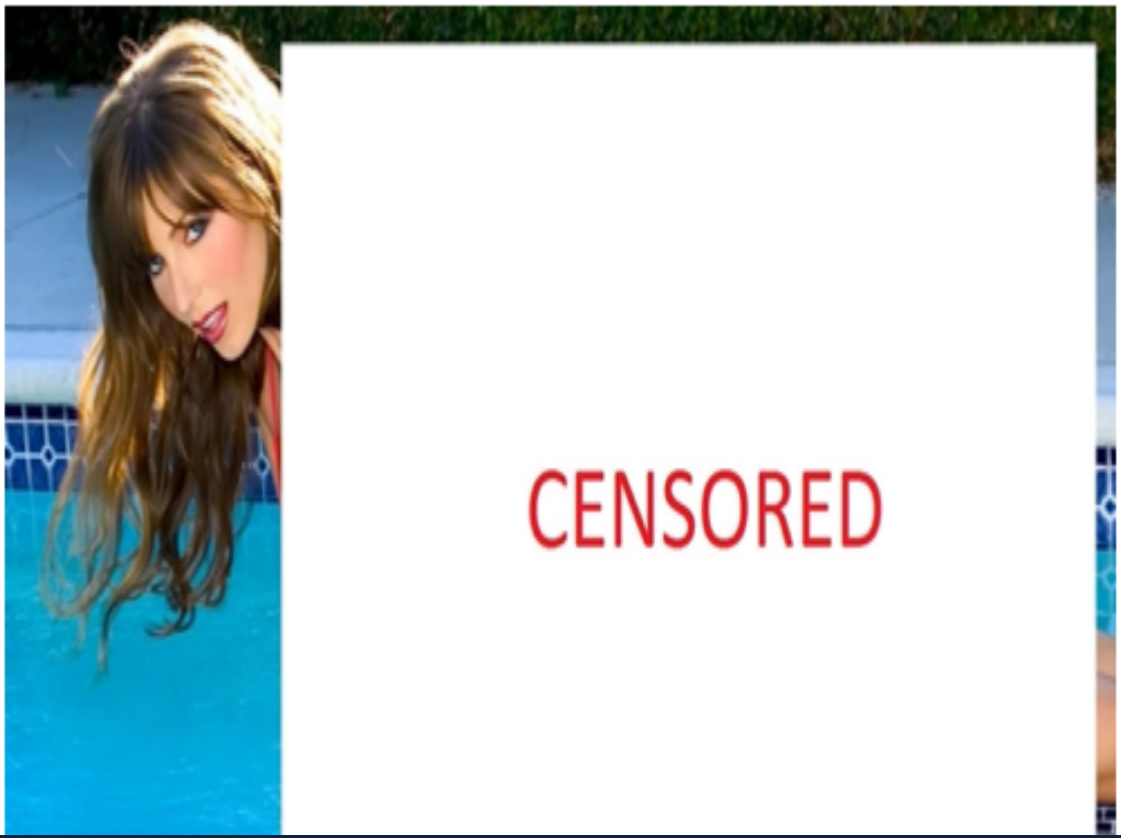
Regression to the mean: Wealth

If your income is at the 98th percentile of the income distribution — that is, you earn more than 98 percent of the population — the best guess is that your children, when they are adults, will be in the 65th percentile.



Regression to the mean: Examples

IN 2006, TRADINGMARKETS, a company that helps people trade stocks, asked ten Playboy Playmates to select five stocks each. The idea was to see if they could beat the market. The winner was Deanna Brooks, Playmate of the Month in May 1998. The stocks she picked rose 43.4 percent, trouncing the S& P 500, which gained 13.6 percent, and beating more than 90 percent of the money managers who actively try to outperform a given index. Brooks wasn't the only one who fared well. Four of the other ten Playmates had better returns than the S& P 500 while less than a third of the active money managers did.

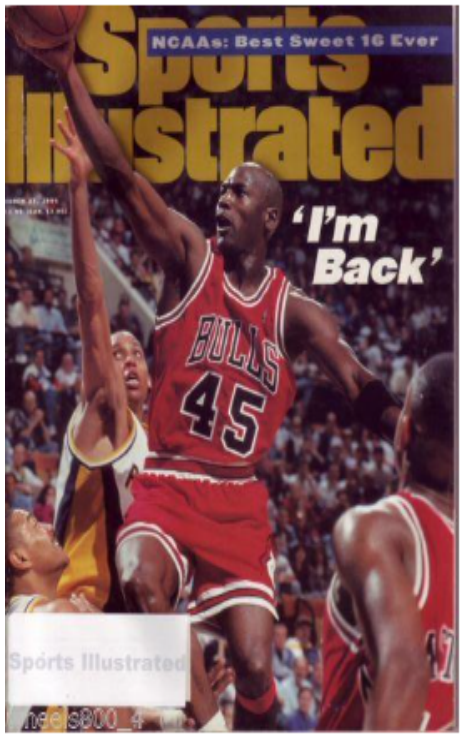


Sports Illustrated cover Jynx



***The Sports Illustrated* cover jinx** is an urban legend that states that individuals or teams who appear on the cover of the Sports Illustrated magazine will subsequently be jinxed (experience bad luck)

Exception



::::

Regression to the mean: Definitions

Regression to the mean is a statistical phenomenon that can make natural variation in repeated data look like real change. It happens when unusually large or small measurements tend to be followed by measurements that are closer to the mean.

Regression to the mean is a technical way of saying that things tend to even out over time.

The term [**Regression to the mean**] is now generally used to label the phenomenon that a variable that is extreme on its first measurement will tend to be closer to the center of the distribution for a later measurement.

True score theory

From the perspective of classical test theory, an examinee's obtained test score (X) is composed of two components, a true score component (T) and an error component (E)

$$X = T + E$$

Skills and Luck: the formula

$$Outcome = Skills + Luck$$

$$Skills = Outcome - Luck$$

How much is the skill and how much is the luck ?

$$Skills = Outcome - Luck$$

$$Var(Skills) = Var(Outcome) - Var(luck)$$

Example: NBA

We will assume that the game outcome has Binomial distribution

$$P(\text{win x games}) = \frac{n!}{x!(n-x)!} p^x * (1-p)^{(n-x)}$$

$$\mu = np$$

$$\sigma^2 = np(1-p)$$

Example: NBA

```
data("nba_misc")
head(nba_misc)
```

```
##   Rk                  Team  Age  W  L PW PL  MOV   SOS  SRS  ORtg  DRtg  Pace
## 1  1       Milwaukee Bucks 26.9 60 22 61 21 8.87 -0.82 8.04 113.8 105.2 103.3
## 2  2  Golden State Warriors 28.4 57 25 56 26 6.46 -0.04 6.42 115.9 109.5 100.9
## 3  3       Toronto Raptors 27.3 58 24 56 26 6.09 -0.60 5.49 113.1 107.1 100.2
## 4  4             Utah Jazz 27.3 50 32 54 28 5.26  0.03 5.28 110.9 105.7 100.3
## 5  5       Houston Rockets 29.2 53 29 53 29 4.77  0.19 4.96 115.5 110.7  97.9
## 6  6 Portland Trail Blazers 26.2 53 29 51 31 4.20  0.24 4.43 114.7 110.5  99.1
##     FTr X3PAr   TSP eFGP_o TOVP_o ORBP_o FT_FGA_o eFGP_d TOVP_d DRBP_d FT_FGA_d
## 1 0.255 0.419 0.583  0.550   12.0   20.8    0.197  0.503   11.5   80.3    0.162
## 2 0.227 0.384 0.596  0.565   12.6   22.5    0.182  0.508   11.7   77.1    0.205
## 3 0.247 0.379 0.579  0.543   12.4   21.9    0.198  0.509   13.1   77.1    0.190
## 4 0.295 0.394 0.572  0.538   13.4   22.9    0.217  0.507   12.4   80.3    0.189
## 5 0.279 0.519 0.581  0.542   12.0   22.8    0.221  0.525   13.4   74.4    0.210
## 6 0.258 0.339 0.568  0.528   12.1   26.6    0.210  0.516   11.0   77.9    0.195
##                     Arena Attendance
## 1            Fiserv Forum     721692
## 2            Oracle Arena     803436
## 3        Scotiabank Arena     812822
## 4 Vivint Smart Home Arena     750546
## 5           Toyota Center     740392
## 6             Moda Center     799345
```

Example: NBA

Every team has played 82 games. Create another variable with win ratio

```
nba_misc$WR<-nba_misc$W/82
```

Lets get the variance of the winning ratio- This is the variance for outcome

```
var(nba_misc$WR)
```

```
## [1] 0.02151839
```

Let us look at the variance for the luck. Here we assume that the chance of winning for each team is the same 0.5.

```
0.5*0.5/82
```

```
## [1] 0.00304878
```

For variance of the luck (proportion p), we use the following formula

$$\frac{p(1-p)}{n}$$

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.02151839 - 0.00304878

Var(Skills) = 0.01846961

The paradox fo Skill:
More skills means luck is more important

Assuming that the variance in luck stays constant

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.02151839 - 0.00304878

Var(Skills) = 0.01846961

This analysis allows us to look at the ratio of variance (luck) to variance (observed) in order to determine the contribution of luck.

$$\frac{0.00304878}{0.01556699} = 19\%$$

Paradox of skill

The less variation in skill there is, the more important luck is.

Skill or luck ?

Manchester United vs Bayern Munich UCL Final

1999

https://youtu.be/YnecJa5satA?t=112

Manchester United vs Bayern Munich, UCL final 1999

Overall
Box-Score Statistics

Statistic	MU	Bayern
Goals Scored	2	1
Total Shots	15	15
Shots on target	9	7
Ball possession	53%	47%
Corner kicks	12	7
Fouls committed	11	10
Offsides	6	8
Yellow cards	0	1
Red cards	0	0

So what ?

The contribution of luck in some professional sports leagues

League	Contribution of luck	Five seasons ended
National Basketball Association	12%	2011
Premier League	31%	2011
Major League Baseball	34%	2011
National Football League	38%	2011
National Hockey League	53%	2012

Source: Analysis by author.

Mauboussin, Michael J.. The Success Equation: Untangling Skill and Luck in Business, Sports, and Investing. Harvard Business Review Press.

Curse of the Bambino



- In 1919 Red Sox sold their star player, Babe Ruth to New York Yankees
- Red Sox was not able to win World Series for the next 86 years, despite of being a favorite in many final games

Regression to the mean

Construct the variable DF of difference

```
data(nba_east)
data(nba_west)
nba_st <- rbind(nba_east,nba_west)


nba_st$PW <- nba_st$PS^14.105/(nba_st$PS^14.105+nba_st$PA^14.105)
nba_st$G <- nba_st$W+nba_st$L
nba_st$PW_G <- round(nba_st$PW*nba_st$G)

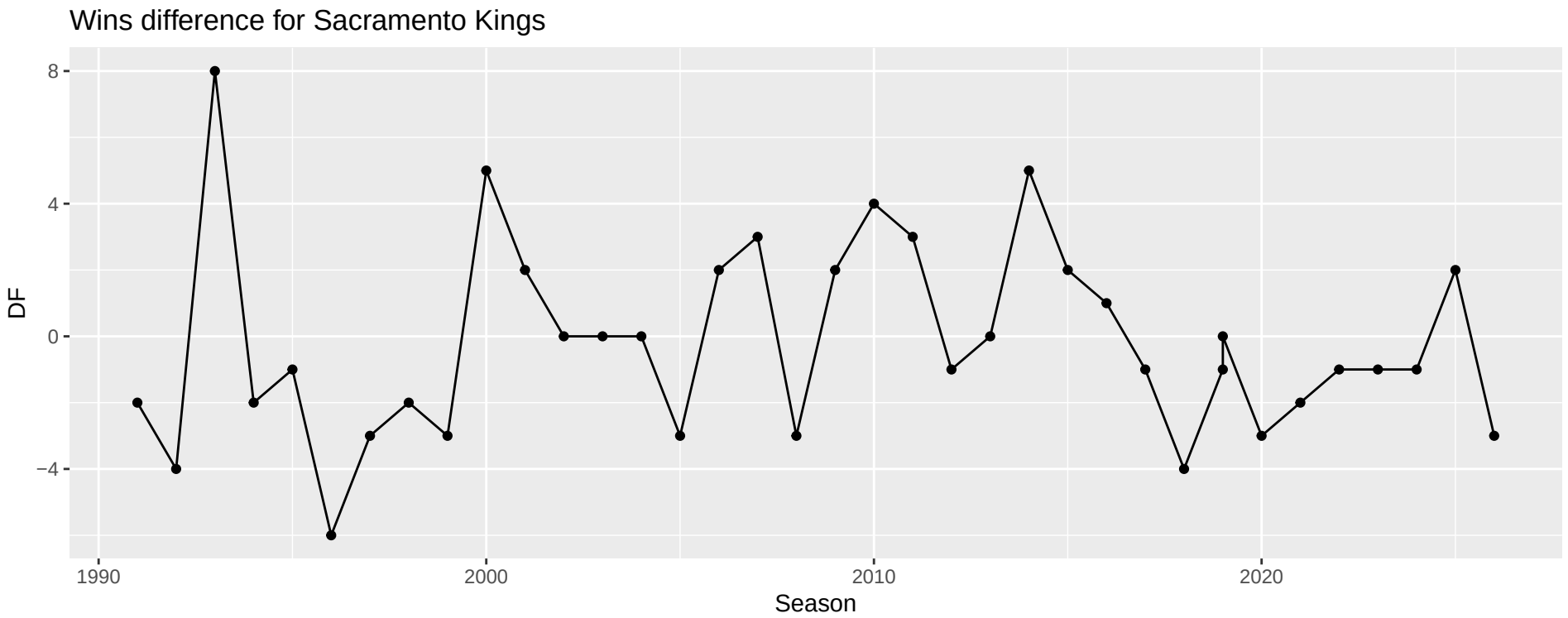
nba_st$DF <- nba_st$PW_G-nba_st$W
```

Regression to the mean

Let's plot for one team

```
nba_st$DF <- nba_st$PW_G-nba_st$W

nba_st %>% filter(Team=="Sacramento Kings") %>%
ggplot(aes(x=Season, y=DF))+geom_point()+geom_line()+
  ggtitle('Wins difference for Sacramento Kings')
```



Regression to the mean

```
means <- nba_st %>% group_by(Team) %>% summarise(mean=mean(DF))
head(means)
```

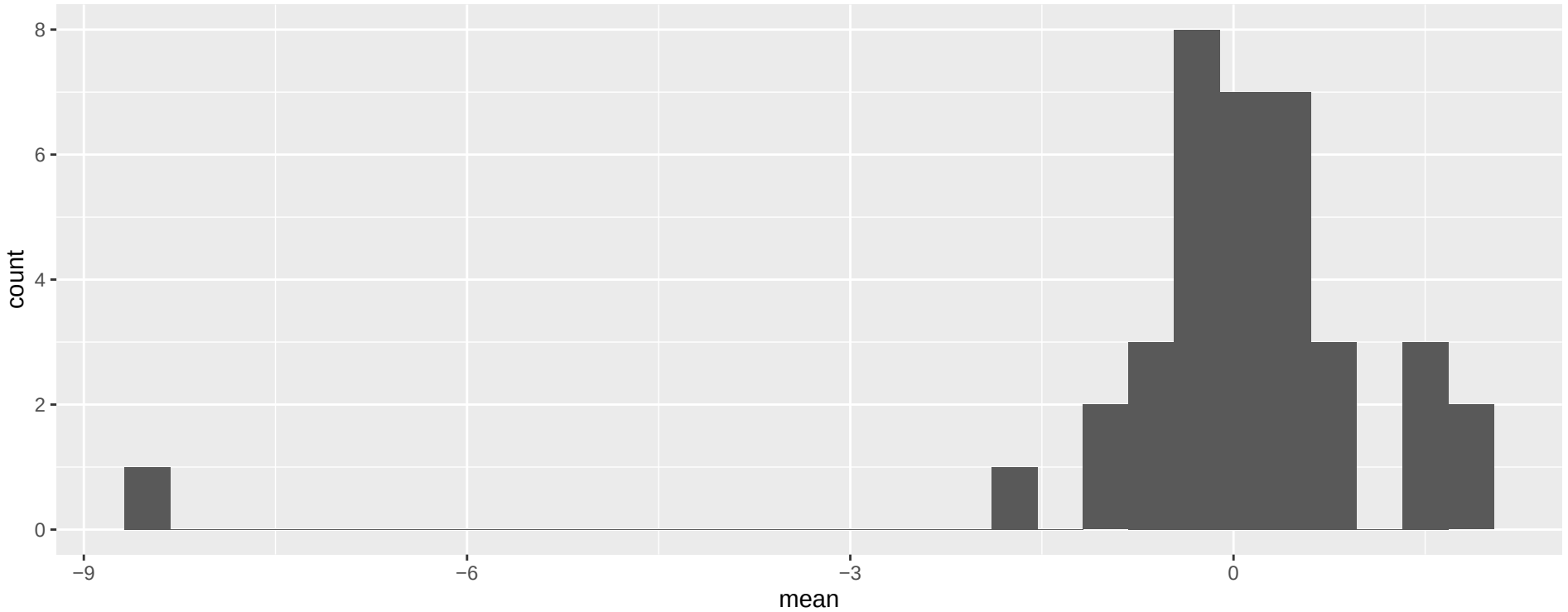
```
## # A tibble: 6 x 2
##   Team                 mean
##   <chr>               <dbl>
## 1 Atlanta Hawks      -0.486
## 2 Boston Celtics      0.784
## 3 Brooklyn Nets      -1.13
## 4 Charlotte Bobcats -0.5
## 5 Charlotte Hornets -0.32
## 6 Chicago Bulls      -0.622
```

Average difference by teams (You need dplyr here)

Regression to the mean

The histogram of the averages

```
ggplot(means, aes(x=mean))+geom_histogram()
```



Competitive Balance in Sports

Competitive Balance in Sports

Competitive Balance

Definition from economics:

A market situation where no business is too big or has an unfair advantage. Source: http://lexicon.ft.com/Term?term=competitive-balance

Definition in Sports:

The degree of uncertainty in the match outcome or the final league standings.

Higher is the Competitive Balance, higher is the uncertainty

Competitive Balance: Dimensions

Long-run

Distribution of championships over time

Seasonal

Within season

Short-run

Match uncertainty Outcome of contest

Measuring Seasonal Dimensions

- Statistical measures
  - Range, variance or deviation
- Standard deviation (SD)
  - Dispersion from mean
  - If SD is low, then CB is high
- CB is often measured by
  - SD of actual wins $\div$ idealized SD
  - As measure increases, competitive balance worsens

Seriea: Measuring balance with standard deviation

```
data("seriea_st")
colnames(seriea_st)
```

```
##  [1] "Rank"   "Team"   "M"      "W"      "D"      "L"      "Goals"  "Diff"
##  [9] "Pt"     "Season"
```

The dataset

```
head(seriea_st)
```

```
##   Rank           Team  M  W  D L Goals Diff Pt Season
## 1    1       AC Milan 34 21 10 3 60:24   36 73   1996
## 2    2       Juventus 34 19  8 7 58:35   23 65   1996
## 3    3     Lazio Roma 34 17  8 9 66:38   28 59   1996
## 4    4 ACF Fiorentina 34 17  8 9 53:41   12 59   1996
## 5    5        AS Roma 34 16 10 8 51:34   17 58   1996
## 6    6       Parma AC 34 16 10 8 44:31   13 58   1996
```

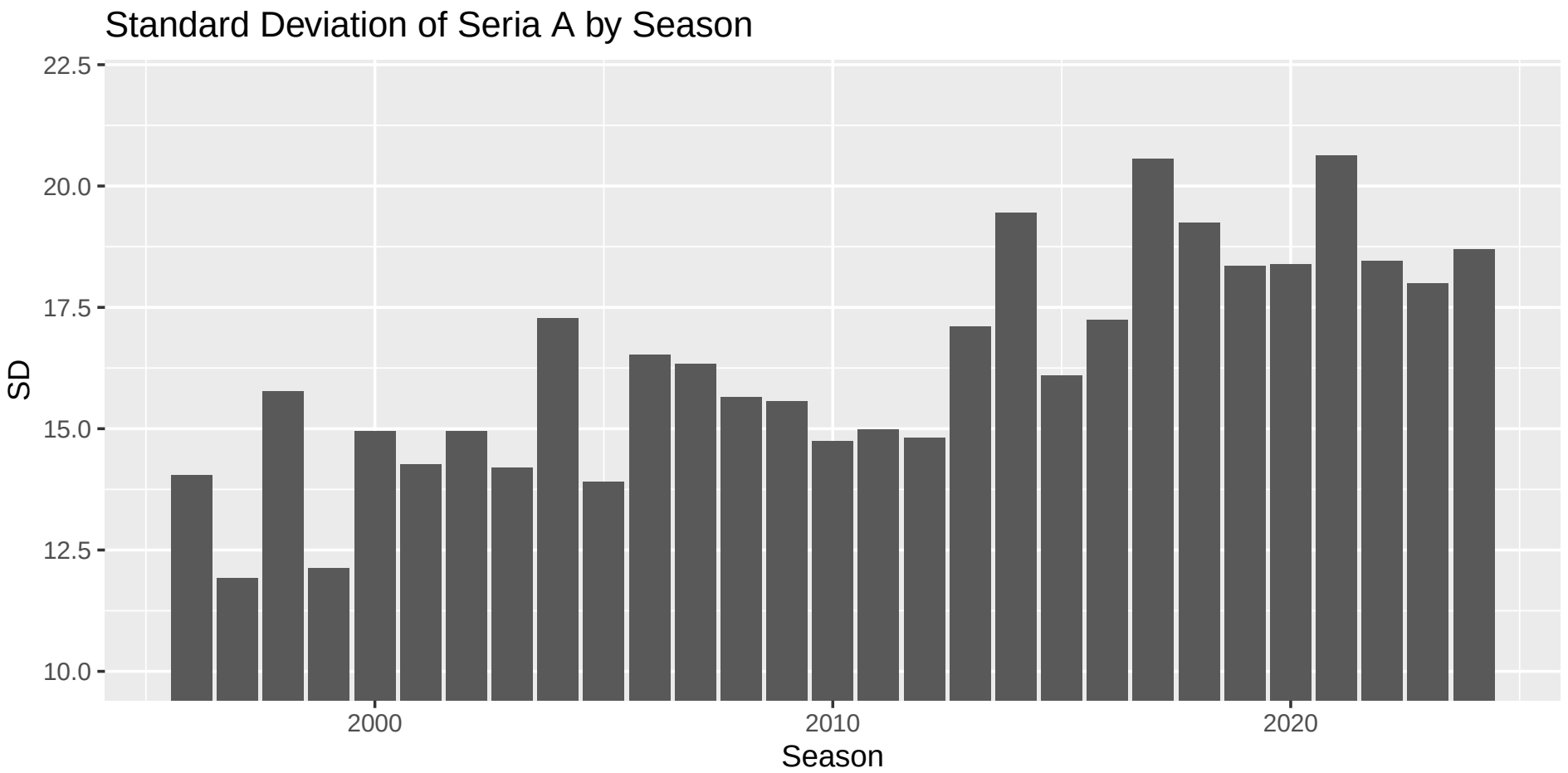
Standard Deviations: Serie A

```
(seriea_s<-seriea_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Pt)))
```

```
## # A tibble: 29 x 2
##    Season    SD
##     <dbl> <dbl>
##  1   1996  14.0
##  2   1997  11.9
##  3   1998  15.8
##  4   1999  12.1
##  5   2000  15.0
##  6   2001  14.3
##  7   2002  14.9
##  8   2003  14.2
##  9   2004  17.3
## 10   2005  13.9
## # i 19 more rows
```

Standard Deviations: Serie A

```
ggplot(seriea_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("Standard Deviation of Seria A by Season")+
  coord_cartesian(ylim=c(10,22))
```



NFL

NFL data

```
data(nfl_st)
summary(nfl_st)
```

```
##       Tm                  W                L                T
##  Length:891         Min.   : 0.000   Min.   : 0.000   Min.   :-1.0000
##  Class :character   1st Qu.: 6.000   1st Qu.: 6.000   1st Qu.: 0.0000
##  Mode  :character   Median : 8.000   Median : 8.000   Median : 0.0000
##                     Mean   : 8.072   Mean   : 8.072   Mean   :-0.1077
##                     3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.: 0.0000
##                     Max.   :16.000   Max.   :16.000   Max.   : 1.0000
##       Wpct              PF              PA              PD
##  Min.   :0.0000   Min.   :161.0   Min.   :165.0   Min.   :-261.0
##  1st Qu.:0.3750   1st Qu.:305.0   1st Qu.:315.0   1st Qu.: -73.5
##  Median :0.5000   Median :354.0   Median :354.0   Median :   1.0
##  Mean   :0.5002   Mean   :356.8   Mean   :356.8   Mean   :   0.0
##  3rd Qu.:0.6250   3rd Qu.:403.5   3rd Qu.:398.0   3rd Qu.:  72.0
##  Max.   :1.0000   Max.   :606.0   Max.   :534.0   Max.   : 315.0
##       MoV                 SoS                SRS
##  Min.   :-1.630e+01   Min.   :-5.900000   Min.   :-17.400000
##  1st Qu.:-4.600e+00   1st Qu.:-1.100000   1st Qu.: -4.300000
##  Median : 1.000e-01   Median : 0.000000   Median :  0.000000
##  Mean   :-5.612e-04   Mean   : 0.001796   Mean   :  0.001235
##  3rd Qu.: 4.500e+00   3rd Qu.: 1.100000   3rd Qu.:  4.400000
##  Max.   : 1.970e+01   Max.   : 4.300000   Max.   : 20.100000
##       OSRS                DSRS              Season       Conf
##  Min.   :-1.230e+01   Min.   :-9.800000   Min.   :1998   AFC:447
##  1st Qu.:-3.200e+00   1st Qu.:-2.300000   1st Qu.:2005   NFC:444
##  Median :-1.000e-01   Median : 0.200000   Median :2012
##  Mean   : 5.612e-04   Mean   :-0.001908   Mean   :2012
##  3rd Qu.: 2.800e+00   3rd Qu.: 2.400000   3rd Qu.:2019
##  Max.   : 1.590e+01   Max.   : 9.800000   Max.   :2025
##       Div
##  Central: 43
##  East   :232
##  North  :192
```

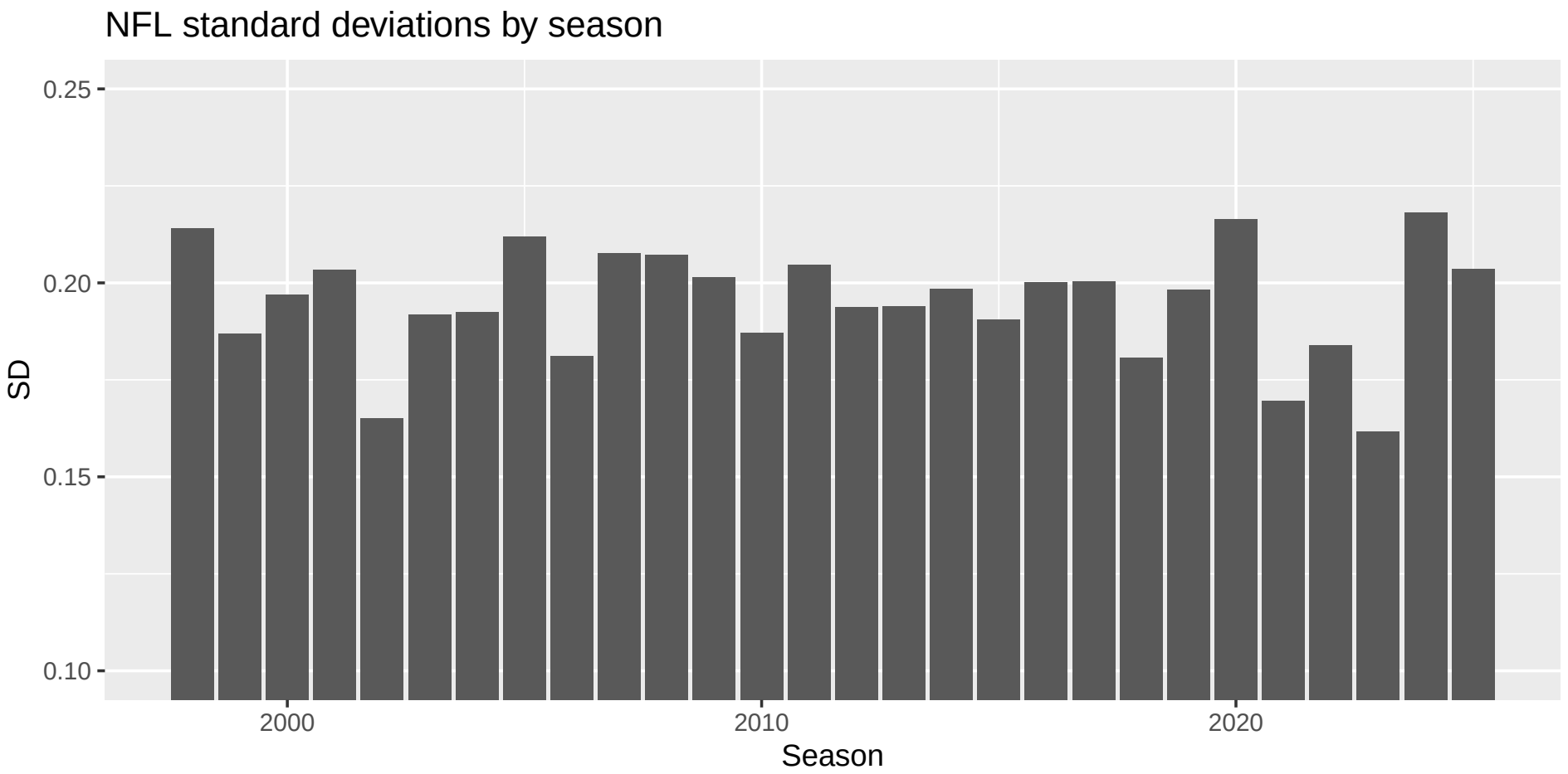
NFL data: SD

```
(nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Wpct)))
```

```
## # A tibble: 28 x 2
##    Season    SD
##     <dbl> <dbl>
##  1   1998 0.214
##  2   1999 0.187
##  3   2000 0.197
##  4   2001 0.203
##  5   2002 0.165
##  6   2003 0.192
##  7   2004 0.192
##  8   2005 0.212
##  9   2006 0.181
## 10   2007 0.208
## # i 18 more rows
```

NFL data: SD

```
ggplot(nfl_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("NFL standard deviations by season")+
  coord_cartesian(ylim=c(0.1,0.25))
```



StDev approach: the drawbacks

- Depends on how the game is played and how the points are distributed
- Cannot be used to compare different sports

Measuring Seasonal Dimensions

There are several drawbacks of using standard deviation as a measure of competitive balance.

The number of the games played are not always constant over time or same for different sports and leagues.

To deal with this problem, σ is benchmarked against standard deviation that describes dispersion for the league with ideal competitive balance.

Measuring Seasonal Dimensions

Lets assume we are dealing with a game, where the match outcome is either win or loose.

As previously, we will define w (wins ratio) as a number of games a team has won, divided by the number of games it has played.

The standard deviation of the wins ratio will be defined as

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (w_i - 0.5)^2}{n}}$$

n - number of teams

w_i - winning ration of team i

Measuring Seasonal Dimensions

Ideally, each team will have 0.5 probability of winning any game (ignoring home-field advantage). Then the ideal standard deviation will be calculated with the following formula

$$\sqrt{\frac{0.5*(1-0.5)}{m}} = \frac{0.5}{\sqrt{m}}$$

Where m is the number of games played by each team.

Measuring Seasonal Dimensions

Now we can define relative measure of competitive balance

$$\sigma_{\text{relative}} = \frac{\sigma}{0.5/\sqrt{m}}$$

This is also called Noll-Scully number

Measuring Seasonal Dimensions: Noll-Scully number

- Ideally, Noll-Scully number will be equal to 1
- Higher is the number, lower is Competitive Balance
- Closer is the number to 1, higher is Competitive Balance

Measuring Seasonal Dimensions: Noll-Scully number

The Noll-Scully compares what a league actually looks like to what it would ideally look like. In a perfect world — i.e. one where actual balance matches the ideal — the metric would return a 1.0, meaning the league is balanced.

If the number is above 1.0 it means the teams are further away in wins than we would expect given the ideal.

If the number is below 1.0 — which is very rare case— it means the teams are closer in wins than we would expect.

Noll-Scully number: Calculations

Noll-Scully number

```
#Ideal StDev
id_s <- 0.5/sqrt(16)
# Win percentage StDev
sd_wpct <- sd(nfl_st$Wpct)
# Noll-Scully number
sd_wpct/id_s
```

```
## [1] 1.540038
```

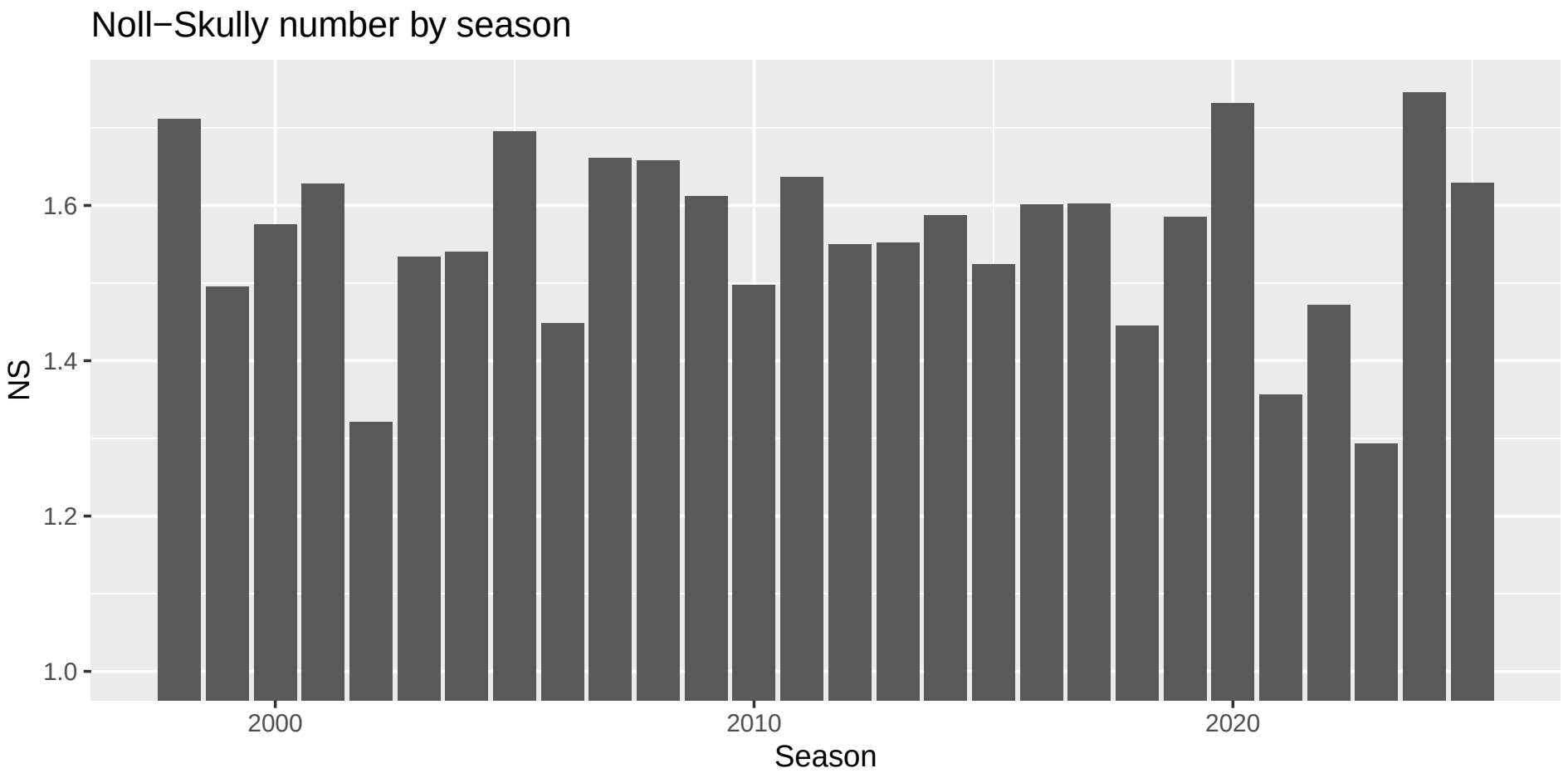
Calculating season-by season

```
(nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(NS=sd(Wpct)/id_s))
```

```
## # A tibble: 28 x 2
##    Season    NS
##     <dbl> <dbl>
##  1   1998  1.71
##  2   1999  1.49
##  3   2000  1.58
##  4   2001  1.63
##  5   2002  1.32
##  6   2003  1.53
##  7   2004  1.54
##  8   2005  1.70
##  9   2006  1.45
## 10   2007  1.66
## # i 18 more rows
```

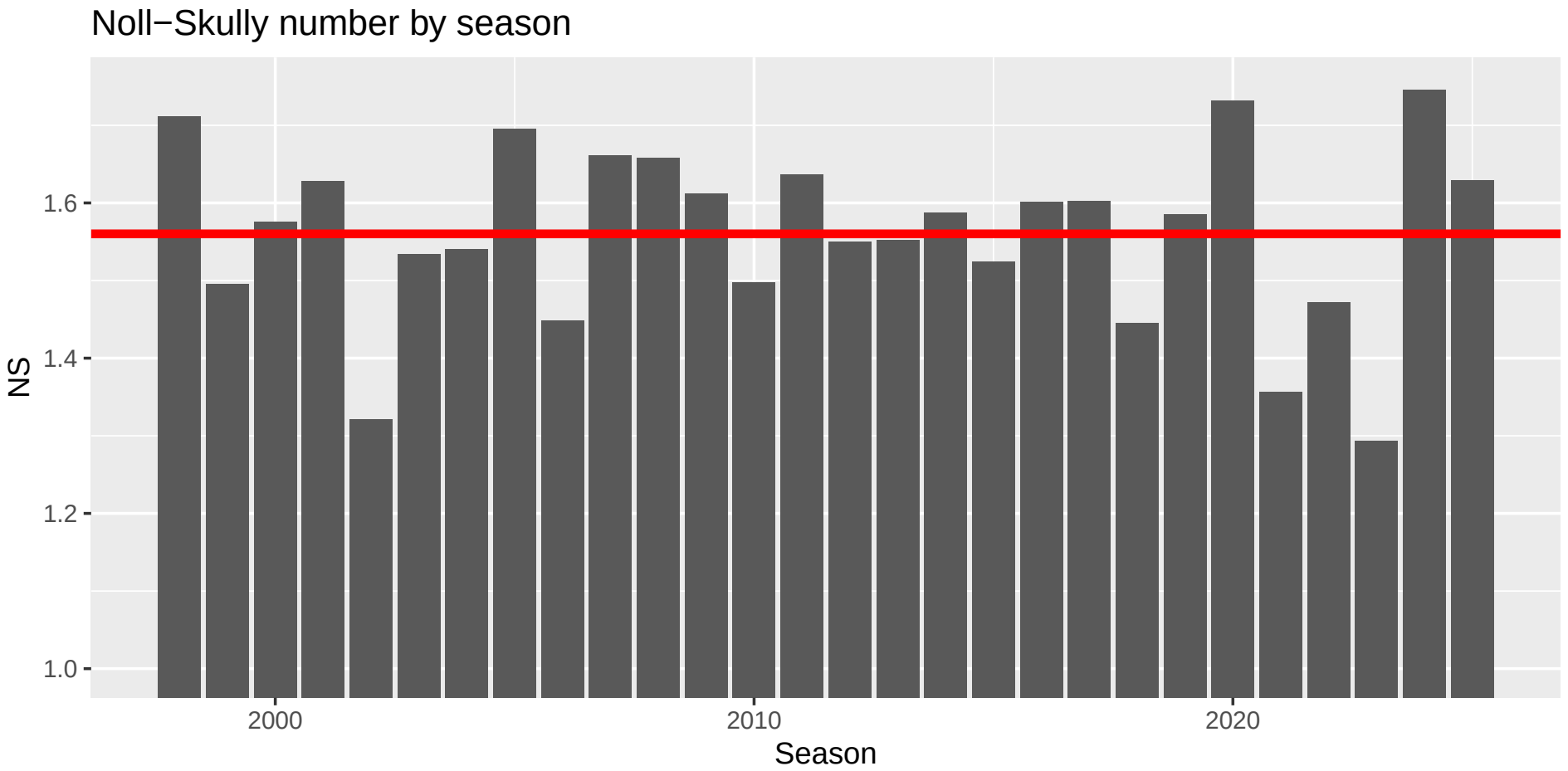
Plotting results

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  coord_cartesian(ylim=c(1,1.75))+
  ggtitle("Noll-Skully number by season")
```



Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  ggtitle("Noll-Skully number by season")+
  coord_cartesian(ylim=c(1,1.75))+
  geom_hline(yintercept=mean(nfl_s$NS), col="red", size=1.5)
```



Previous studies

Twenty Years of Competitive Balance in North America

North American Sports League	Years	Average Noll-Scully Measure of Competitive Balance
National Basketball Association	1985–86 to 2004–05	2.86
American League	1986–2005	1.78
National League	1986–2005	1.76
National Hockey League	1984–85 to 2003–04	1.70
National Football League	1986–2005	1.49

Source: **Berri, D. J., Schmidt, M. B., & Brook, S. L. (2008). *The wages of wins: taking measure of the many myths in modern sport*. Stanford, CA: Stanford University Press.**

Concentration indexes

Concentration indexes

Measuring Seasonal dimensions: Concentration of points

Concentration Index for points

$$C5 = \frac{\text{Number of points earned by top N clubs}}{\text{Total number of points earned by all clubs}}$$

Measuring Seasonal dimensions: Concentration of points

Concentration index for points

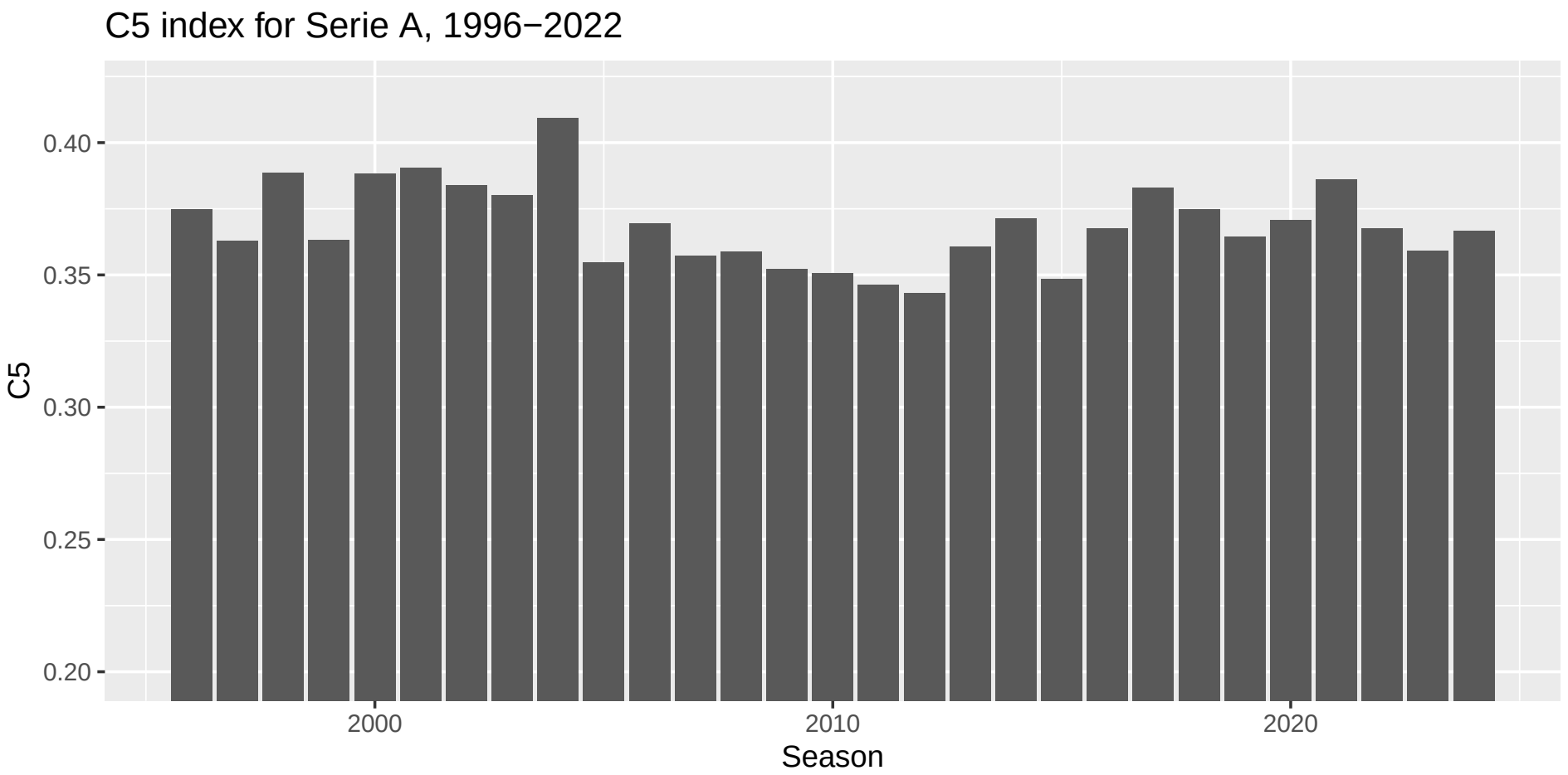
```
top5<- seriea_st %>%
    group_by(Season) %>%
    filter(Rank<6) %>%
    summarize(TopP=sum(Pt))

all<-seriea_st %>%
    group_by(Season) %>%
    summarize(Pt=sum(Pt))

C5<-data.frame(Season=top5$Season,C5=top5$TopP/all$Pt)
```

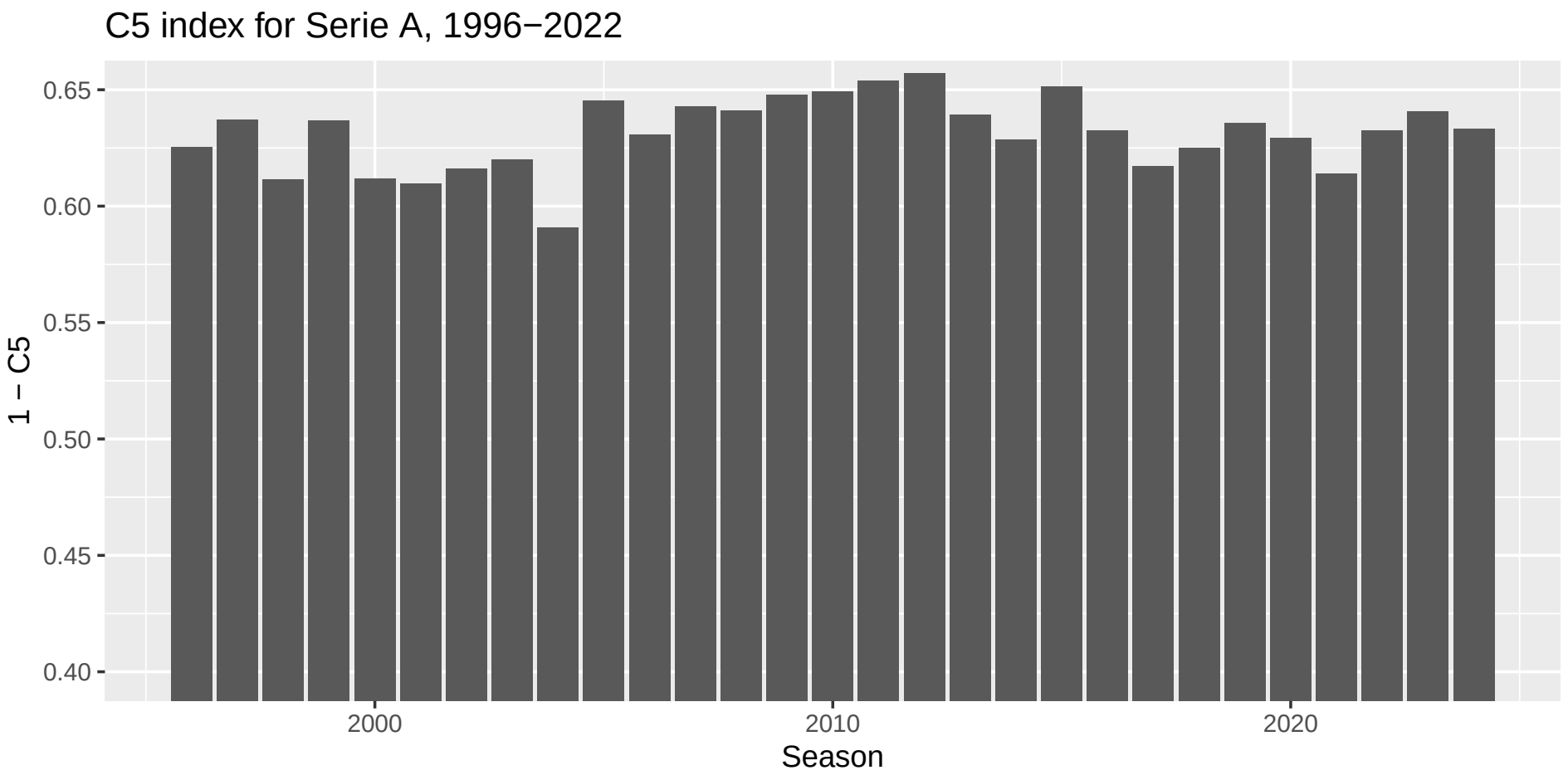
Measuring Seasonal dimensions: Competitive imbalance

```
ggplot(C5, aes(Season, C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2022")+
  coord_cartesian(ylim=c(0.2,0.42))
```



Measuring Seasonal dimensions: Competitive Balance

```
ggplot(C5, aes(Season, 1-C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2022")+
  coord_cartesian(ylim=c(0.4,0.65))
```



Measuring Long Run dimensions

Concentration Index

Proportion of championships (or points) by top n teams

e.g. CI_5 : dominance of top 5 teams

Measuring Long Run dimensions

How many times did every team become a champion?

```
(champ_c <- seriea_st %>%
  filter(Rank ==1) %>% group_by(Team) %>%
  summarize(champ=n()) %>% arrange(desc(champ)))
```

```
## # A tibble: 6 x 2
##   Team        champ
##   <chr>       <int>
## 1 Juventus       16
## 2 Inter           6
## 3 AC Milan        5
## 4 AS Roma         1
## 5 Lazio Roma      1
## 6 SSC Napoli      1
```

CI4 for championship

```
100*sum(champ_c[1:4,2])/sum(champ_c$champ)
```

```
## [1] 93.33333
```

- Juventus won 16 of these titles
- The top four clubs won about 93 percent
- A high championship CI means long run balance is low

Herfindahl-Hirschman Index - HHI

Herfindahl-Hirschman Index - HHI

Herfindahl-Hirschman Index - HHI

Definition comes from economics

The Herfindahl-Hirschman index (HHI) is a commonly accepted measure of market concentration. It is calculated by squaring the market share of each firm competing in a market, and then summing the resulting numbers

Read more: http://www.investopedia.com/terms/h/hhi.asp#ixzz4t8sTjDns

Herfindahl-Hirschman Index - HHI

$$HHI = \sum_{i=1}^{n} w_i^2$$

Simple case: 2 firms in the market

- Highly competitive industry, both firms have the same market share

$HHI = \sum_{i=1}^{2} w_i^2 = 0.5^2 + 0.5^2$

- Monopoly, 1 firm has 100%

$HHI = \sum_{i=1}^{2} w_i^2 = 1^2 + 0^2$

Herfindahl-Hirschman Index - HHI

$$HHI = \sum_{i=1}^{n} w_i^2$$

lower and upper bounds

$$1/n \leq HHI \leq 1$$

Herfindahl-Hirschman Index – in sports

Several variations of the index exists.

$$HHI = \sum_{i=1}^{n} \left(\frac{p_i}{\sum_{i=1}^{n} p_i} \right)^2$$

p_i is the number of the points the team has earned.

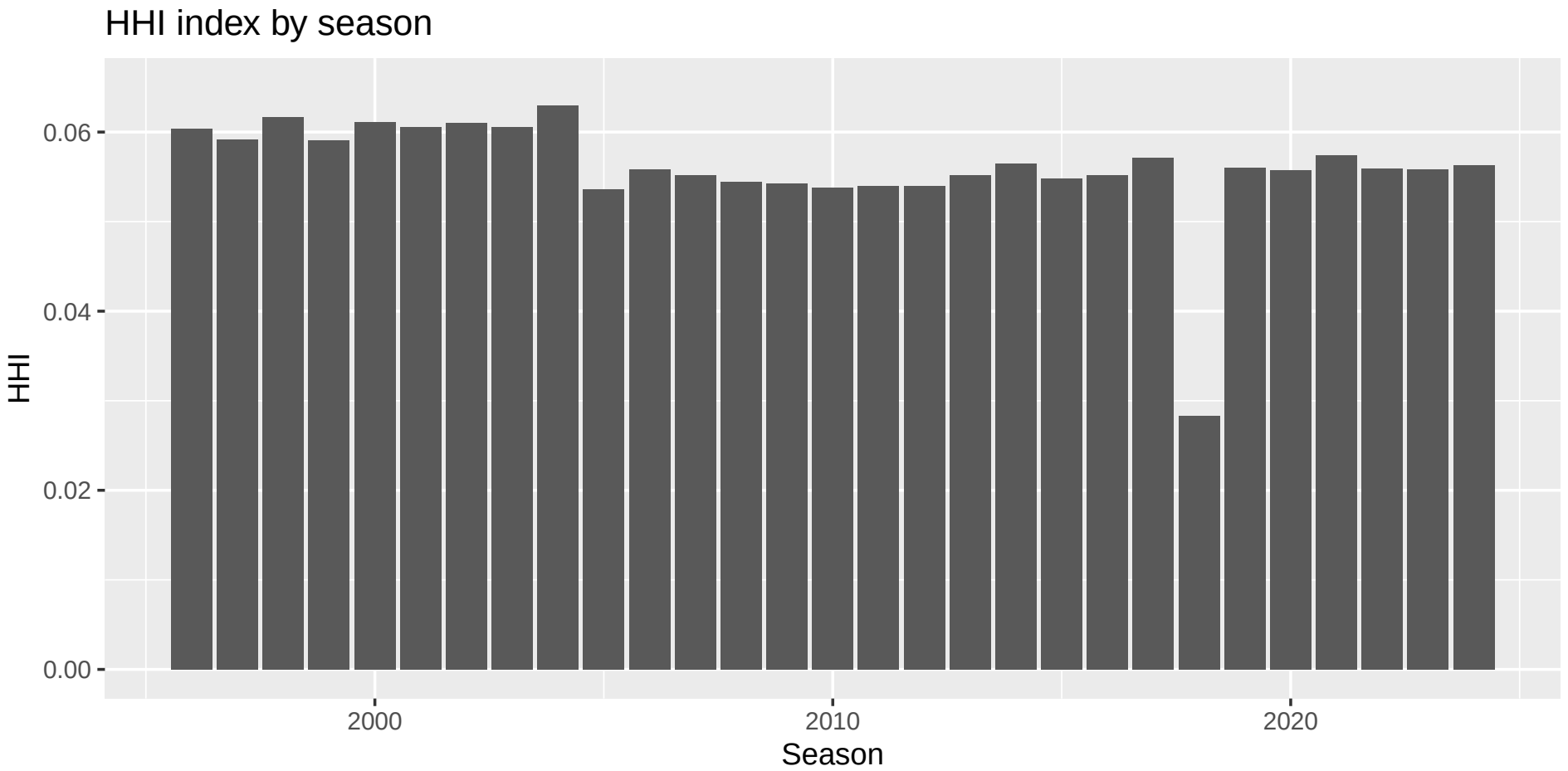
- $p_i / \sum p_i$ is team i's share of all points that season
- Square the shares, then add them
- Every team level on points: each share is $1/n$, so $HHI = 1/n$
- One team takes every point: $HHI = 1$
- A lower HHI means the points are more spread out

Doing it in R: Serie A

```
HHI = seriea_st %>%
        group_by(Season) %>%
        mutate(Perc=Pt/sum(Pt))%>%
        summarise(HHI = sum(Perc^2))
```

Doing it in R: Serie A

```
ggplot(HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  ggtitle("HHI index by season")+
  coord_cartesian(ylim=c(0,0.065))
```



Doing it in R: Serie A

Number of the teams per season

```
seriea_st %>%
      group_by(Season) %>%
      summarise(count=n())
```

```
## # A tibble: 29 x 2
##    Season count
##     <dbl> <int>
##  1   1996    18
##  2   1997    18
##  3   1998    18
##  4   1999    18
##  5   2000    18
##  6   2001    18
##  7   2002    18
##  8   2003    18
##  9   2004    18
## 10   2005    20
## # i 19 more rows
```

- 1996-2004: 18 teams, so equal points means $HHI = 1/18 \approx 0.056$
- From 2005: 20 teams, so equal points means $HHI = 1/20 = 0.05$
- A smaller raw HHI can be the extra clubs, not a closer season
- 2018 has 40 rows in this file. Do not read that by as a real change in balance

Adjusted HHI

Increasing number of the teams affects lower bound (in some cases upper bound as well)

$$dHHI = \sum_{i=1}^{n} w_i^2 - 1/n$$

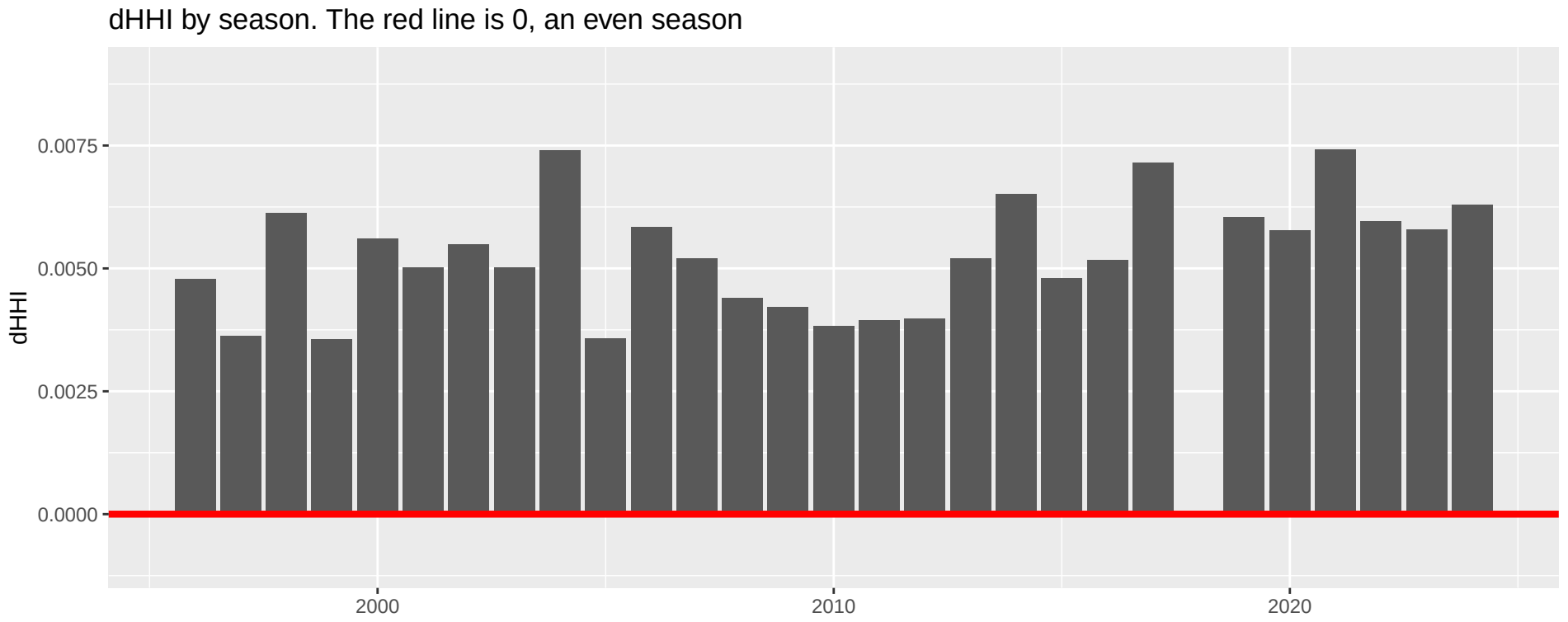
- $1/n$ is the HHI of a perfectly even season
- Subtract it, and a perfectly even season scores 0
- A lower dHHI means a more balanced season
- Use it when n changes. Raw HHI does not survive the move from 18 teams to 20

Adjusted HHI

```
dHHI<-seriea_st %>%
      group_by(Season) %>%
      mutate(Perc=Pt/sum(Pt)) %>%
      summarise(HHI=(sum(Perc^2)-1/n()))
```

dHHI

```
ggplot(dHHI %>% filter(Season != 2018), aes(Season, HHI)) +
  geom_col() +
  geom_hline(yintercept = 0, col = "red", size = 1.5) +
  coord_cartesian(ylim = c(-0.001, 0.009)) +
  labs(y = "dHHI",
       title = "dHHI by season. The red line is 0, an even season")
```



HHI defined by Depken

$$HHI = \sum_{i=1}^{n} \left(\frac{2 * wins_i}{N * G_i}\right)^2$$

$wins_i$- *number of wins by each team*

N - *Number of teams*

G_i - *number of games played*

Assumptions

- No ties
- Number of games played by each is an even number

Source: Free-Agency and the Competitiveness of Major League Baseball, CRAIG A. DEPKEN, II, Review of Industrial Organization, Vol. 14, No. 3 (May 1999), pp. 205-217

HHI defined by Depken

$$HHI = \frac{4}{N^2 G^2} \sum_{i=1}^{n} (wins_i)^2$$

Source: Free-Agency and the Competitiveness of Major League Baseball, CRAIG A. DEPKEN, II, Review of Industrial Organization, Vol. 14, No. 3 (May 1999), pp. 205-217

- Each game produces one win, so the league's wins total $NG/2$
- $2 \times wins_i/(NG)$ is team i's share of those wins
- Every team wins half its games: $HHI = 1/N$
- For 30 MLB clubs that floor is $1/30 \approx 0.033$. That is the red line
- The formula needs the same G for every club. The 2020 season was 60 games, so leaving 162 in the formula makes 2020 look far too balanced

Major League of Baseball: dataset

```
str(mlb_standings)
```

```
## 'data.frame':    810 obs. of  10 variables:
##  $ Season: int  1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 ...
##  $ Lig   : chr  "AL" "AL" "AL" "AL" ...
##  $ Div   : chr  "W" "W" "W" "W" ...
##  $ Team  : chr  "Anaheim Angels" "Anaheim Angels" "Anaheim Angels" "Anaheim Angel
##  $ W     : int  70 82 75 99 77 92 95 89 94 100 ...
##  $ L     : int  92 80 87 63 85 70 67 73 68 62 ...
##  $ G     : int  162 162 162 162 162 162 162 162 162 162 ...
##  $ R     : int  711 864 691 851 736 836 761 766 822 765 ...
##  $ RA    : int  826 869 730 644 743 734 643 732 731 697 ...
##  $ Rank  : int  4 3 3 2 3 1 1 2 1 1 ...
```

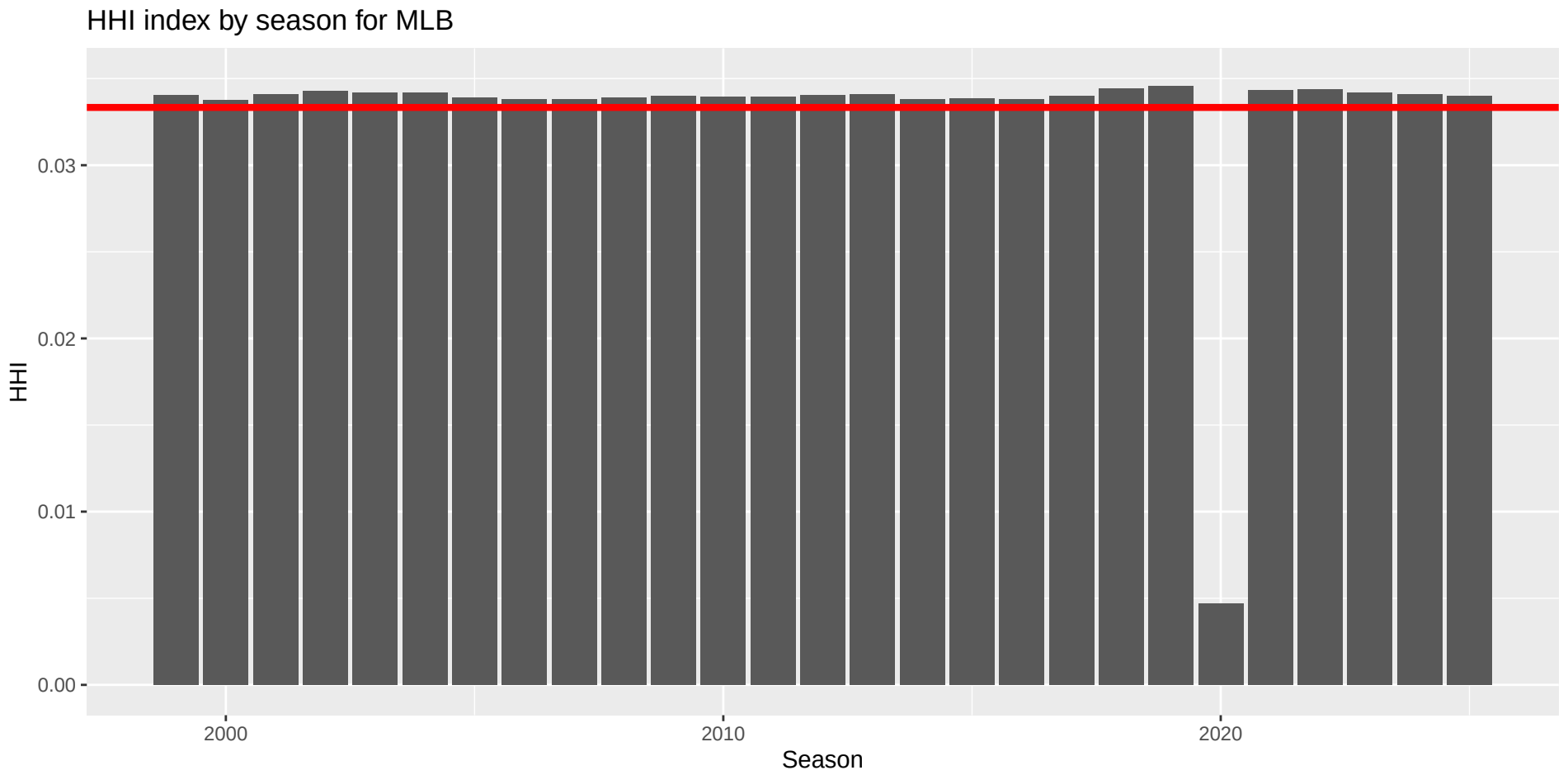
MLB

- Each team plays 162 games per regular season
  - small violations occur, 160,161,163
  - We will assume the number of games to be 162
- 30 teams since 1999

```
mlb_HHI <- mlb_standings %>%
      group_by(Season) %>%
      summarise(HHI=sum(W^2)*(4/(30^2*162^2)))
```

HHI in MLB

```
ggplot(mlb_HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  geom_hline(yintercept = 1/30, col="red", size=1.5)+
  coord_cartesian(ylim=c(0,0.035))+
  ggtitle("HHI index by season for MLB")
```



Correlating imbalance measures with other variables

- Leading indicators: input indicators
  Why is there a imbalance ?
  Salary gap ? Macro, socio-economic variables?
- Lagging indicators: output indicators
  - What is affected by the change in imbalance ?
  - Attendance?
  - Teams revenue ?

MLB: Salaries

Leading indicators: SalarY

```
data(mlb_salary)
str(mlb_salary)
```

```
## 'data.frame':    938 obs. of  6 variables:
##  $ Team       :List of 938
##   ..$ : chr "New York Yankees"
##   ..$ : chr "Texas Rangers"
##   ..$ : chr "Atlanta Braves"
##   ..$ : chr "Cleveland Indians"
##   ..$ : chr "Boston Red Sox"
##   ..$ : chr "New York Mets"
##   ..$ : chr "Los Angeles Dodgers"
##   ..$ : chr "Baltimore Orioles"
##   ..$ : chr "Arizona Diamondbacks"
##   ..$ : chr "Chicago Cubs"
##   ..$ : chr "Houston Astros"
##   ..$ : chr "Colorado Rockies"
##   ..$ : chr "Los Angeles Angels of Anaheim"
##   ..$ : chr "Toronto Blue Jays"
##   ..$ : chr "St. Louis Cardinals"
##   ..$ : chr "San Francisco Giants"
##   ..$ : chr "San Diego Padres"
```

MLB: Salaries

Aggregate data by year

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current))
```

```
both <- inner_join(mlb_s, mlb_HHI, by = "Season")
cor(both$Tot.salary, both$HHI)
```

```
## [1] 0.1127333
```

MLB: Salaries

What about the standard deviation of the salary ?

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current), SD=sd(Current))
```

```
both <- inner_join(mlb_s, mlb_HHI, by = "Season")
cor(both$SD, both$HHI)
```

```
## [1] 0.2726654
```

MLB: Salaries

HHI for salary

```
mlb_HHI_s<-mlb_salary %>%
        group_by(Season) %>%
        mutate(Perc=Current/sum(Current)) %>%
        summarise(HHI_S=sum(Perc^2))
```

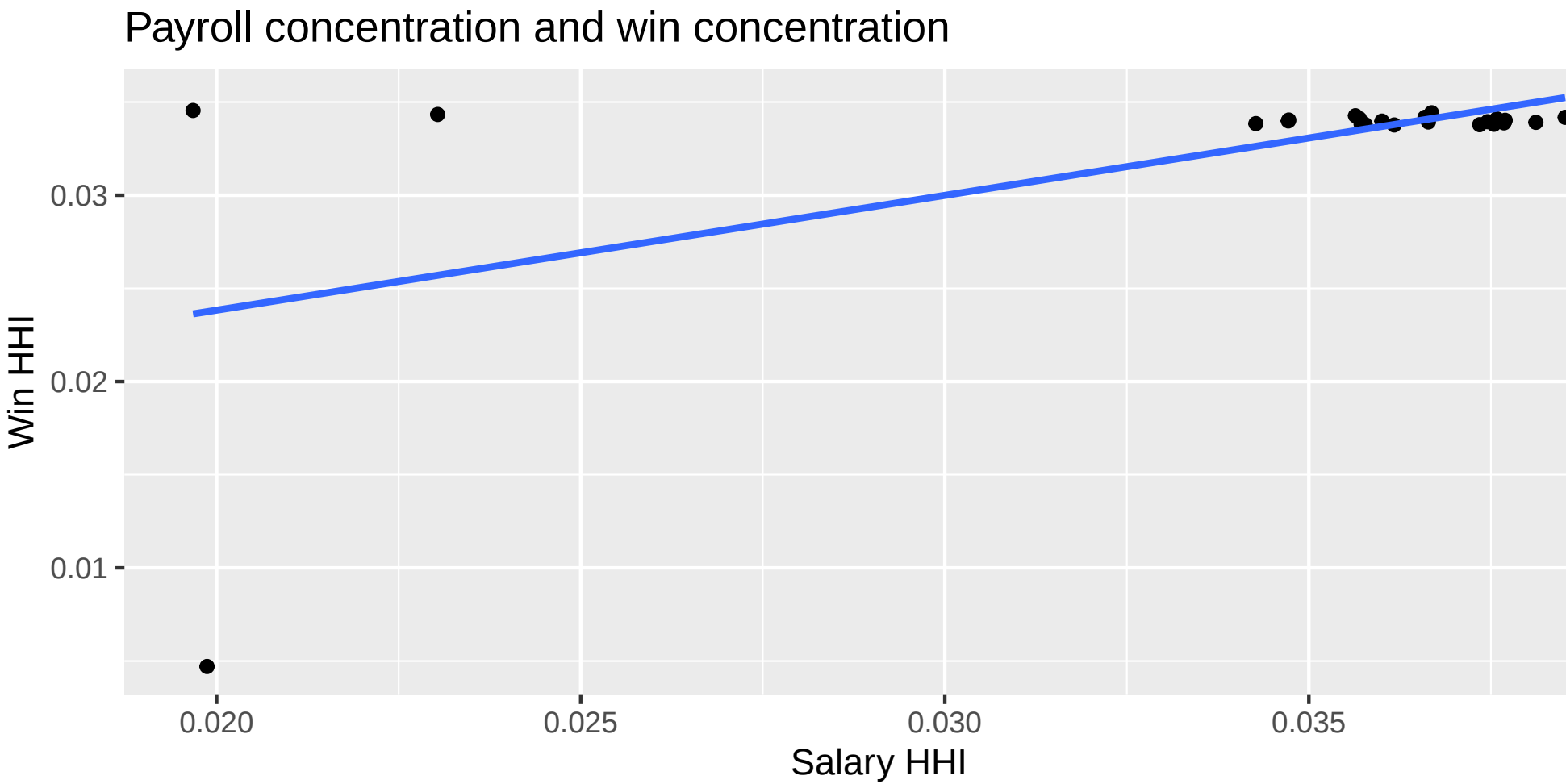
Correlations for HHI_S and HHI

```
both <- inner_join(mlb_HHI_s, mlb_HHI, by = "Season")
cor(both$HHI_S, both$HHI)
```

```
## [1] 0.5578874
```

- Total payroll rises over time, so its correlation with win HHI is mostly that trend
- Salary HHI asks whether the money and the wins concentrated in the same seasons

MLB: Salaries



Short run competitiveness

Short run competitiveness

Short run competitiveness

- We are looking at how predictable are individual games.
- We can do it by using some machine learning or simpler algorithms (like Elo) to predict the probability of game outcome then calculate the prediction error – High error will indicate higher uncertainty, thus higher competitiveness.
- However, here we will look at the bookmaker's odds as a predicted probabilities of events

Match Outcome uncertainty

- Filter the data, by taking seasons starting from 2009, major leagues and exclude NA's
- The variables H,D,A are bookmaker's odds from Bet365

```
data("f_data_sm")
f_2009 <- f_data_sm %>%
  filter(SEASON > 2008,
         COUNTRY %in% c("Italy", "Germany", "Spain","England",
            "Netherlands", "Portugal"), !is.na(H) )
head(f_2009)
```

```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG    H   D     A
## 1    1    0    1 1.20 6.5 15.00
## 2    3    1    4 1.83 3.5  4.50
## 3    2    3    5 2.00 3.3  4.00
## 4    2    1    3 2.60 3.2  2.80
## 5    2    1    3 3.20 3.4  2.25
## 6    0    1    1 5.50 3.6  1.67
```

Match Outcome uncertainty

- We need to convert odds to probabilities
- We also need vig-free probabilities, so the three chances sum to 1
- The vig is the bookmaker's margin built into the odds
- Try library odds.converter

```
( x <- odds.fv(c(1.4,3.8,8), input = 'dec', output = 'prob'))
```

```
## [1] 0.6479113 0.2387042 0.1133845
```

```
sum(x)
```

```
## [1] 1
```

Match Outcome uncertainty

Brier score. Average squared distance between the probabilities and what happened.

$$BS = \frac{1}{M} \sum_{m=1}^{M} \sum_{k=1}^{3} (p_{mk} - o_{mk})^2$$

- p_{mk}: vig-free probability of home, draw, or away
- o_{mk}: 1 if that result happened, 0 if it did not
- A result that was certain, and then happens, scores 0
- A match called 1/3, 1/3, 1/3 scores $2/3 \approx 0.67$
- A heavy favorite, 0.8 / 0.1 / 0.1, scores 0.06 if that favorite wins
- A higher Brier score means the matches were harder to call

Match Outcome uncertainty

Replace variables H,D,A with vig-free probabilities

```
f_2009[,c("H", "D", "A")] <- odds.fv(f_2009[,c("H", "D", "A")],
                        input = 'dec',output = 'prob')
head(f_2009)
```

```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG         H         D         A
## 1    1    0    1 0.7907543 0.1459854 0.06326034
## 2    3    1    4 0.5182626 0.2709773 0.21076012
## 3    2    3    5 0.4748201 0.2877698 0.23741007
## 4    2    1    3 0.3648208 0.2964169 0.33876221
## 5    2    1    3 0.2973183 0.2798290 0.42285270
## 6    0    1    1 0.1717862 0.2624511 0.56576278
```

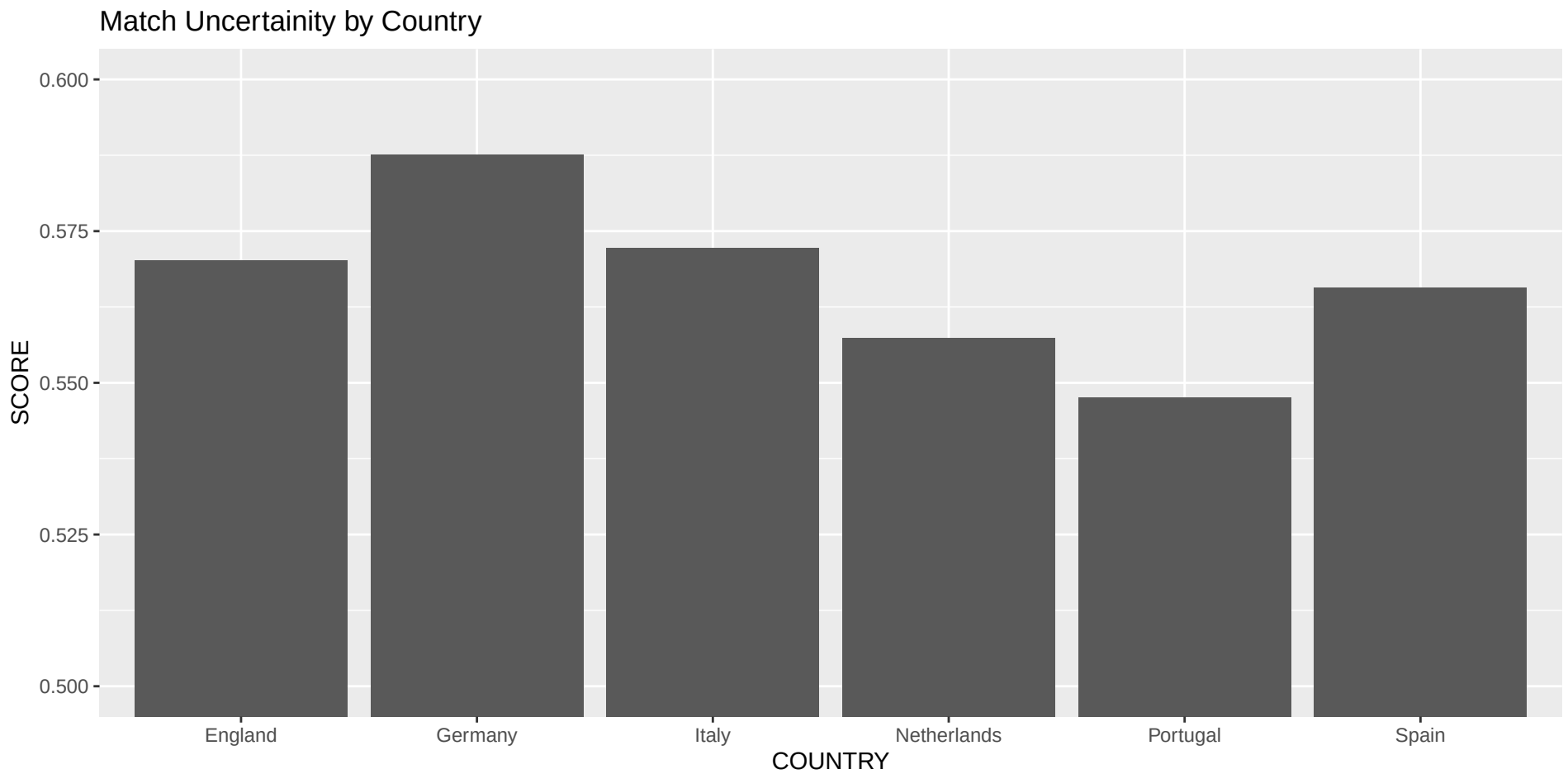
Match Outcome uncertainty

Calculate brier score by country

```
brier_score <- f_2009 %>% group_by(COUNTRY) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
brier_score$SCORE <- unlist(brier_score$SCORE)
```

Match Outcome uncertainty

```
ggplot(brier_score, aes(x=COUNTRY, y = SCORE))+geom_bar(stat='identity')+
  coord_cartesian(ylim = c(0.5,0.6))+ggtitle('Match Uncertainity by Country')
```

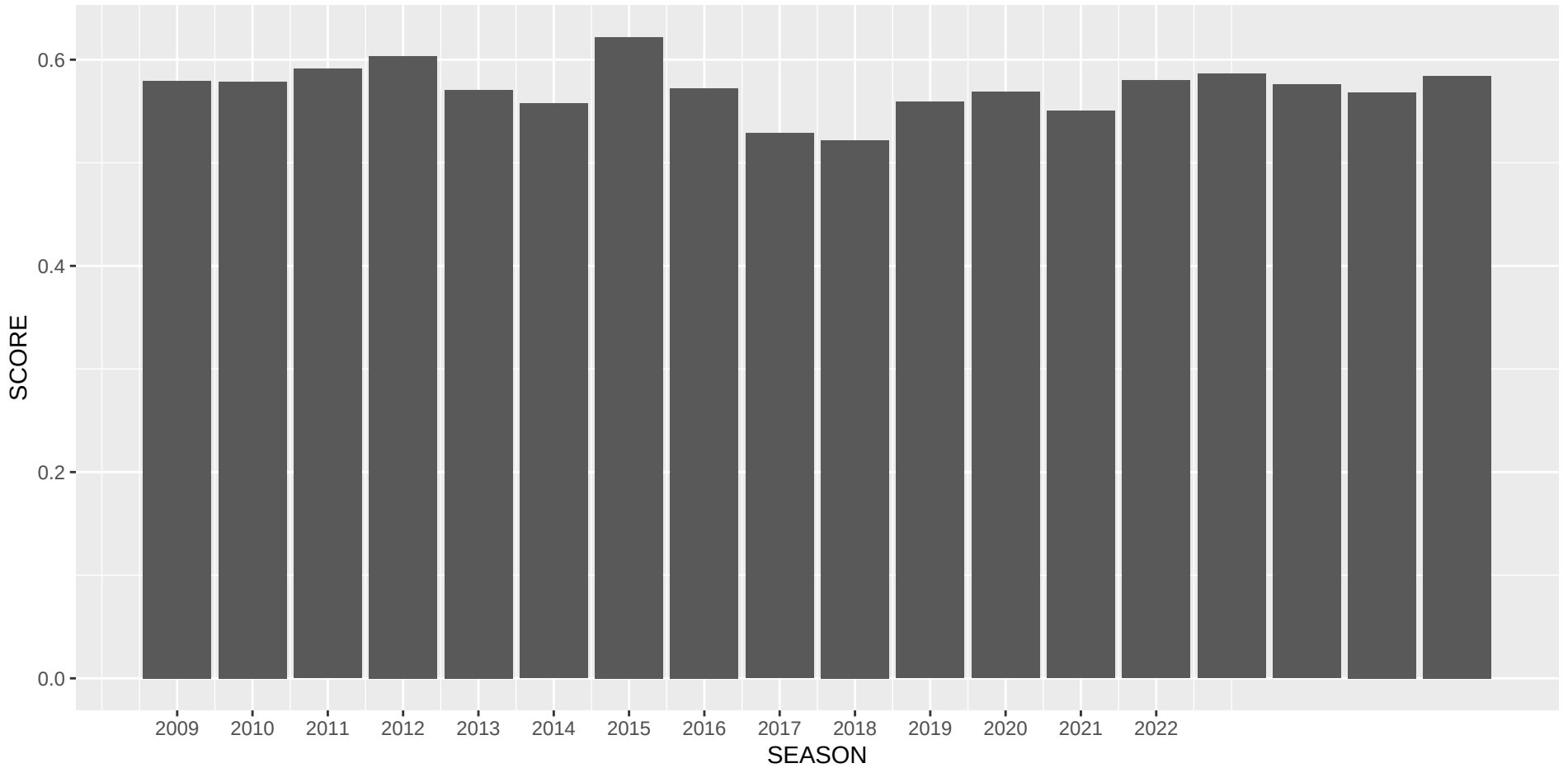


Match Outcome uncertainty

```
brier_score <- f_2009 %>% group_by(COUNTRY, SEASON) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
brier_score$SCORE <- unlist(brier_score$SCORE)
```

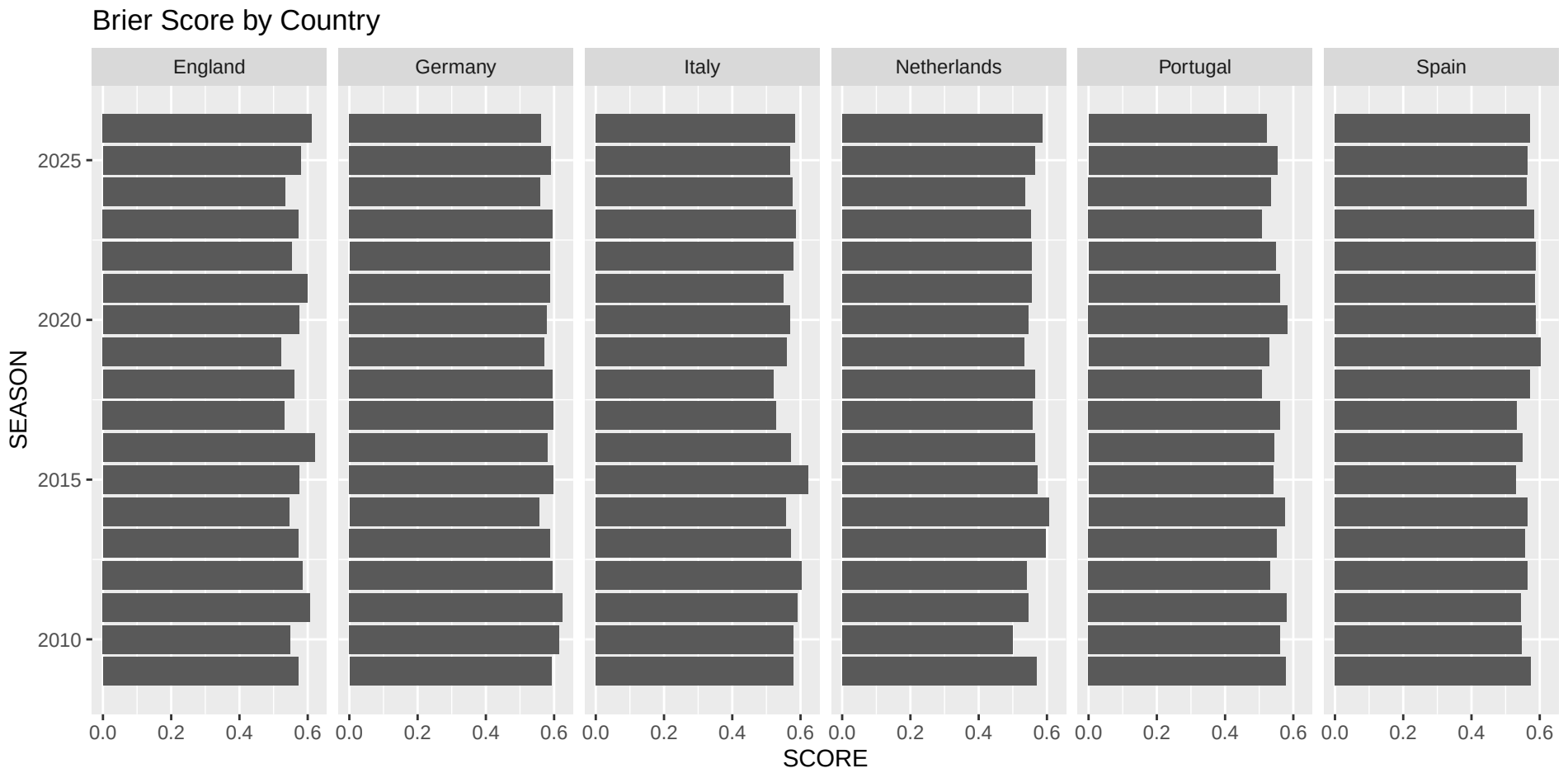
Match Outcome uncertainty: Italy

```
brier_score %>% filter(COUNTRY=='Italy') %>%
ggplot(aes(x=SEASON, y=SCORE))+geom_bar(stat='identity') +
  scale_x_continuous(breaks = 2009:2022)
```



Match Outcome uncertainty

```
ggplot(brier_score, aes(x=SEASON, y=SCORE))+
  geom_bar(stat='identity',position=position_dodge(width=1))+
  facet_grid(~COUNTRY)+ coord_flip()+ggtitle("Brier Score by Country")
```



How to read the numbers

Read the pair. $\downarrow\downarrow$ means a lower number is less competitive. $\downarrow\uparrow$ means a lower number is more competitive.

Metric	Number	Competitiveness
SD of points or win percent	$\downarrow$	$\uparrow$
Noll-Scully	$\downarrow$	$\uparrow$
C5, top-5 share of points	$\downarrow$	$\uparrow$
Championship CI	$\downarrow$	$\uparrow$
1 minus C5	$\downarrow$	$\downarrow$

Ideal SD, $0.5/\sqrt{m}$, is not a competitiveness score. A shorter season makes it larger.

How to read the numbers

Read the pair. $\downarrow\downarrow$ means a lower number is less competitive. $\downarrow\uparrow$ means a lower number is more competitive.

Metric	Number	Competitiveness
HHI	$\downarrow$	$\uparrow$
dHHI	$\downarrow$	$\uparrow$
Depken HHI	$\downarrow$	$\uparrow$
Brier score	$\downarrow$	$\downarrow$
Salary HHI	$\downarrow$	$\uparrow$

Salary HHI is payroll equality, not the league table. A lower salary HHI means the money is more spread out.